

Applied Predictive Modeling — Chapter 6 Solutions

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1 Chapter 6 in one paragraph

Chapter 6 studies models that are linear in the parameters, emphasizing three recurring ideas: (1) ordinary least squares is simple and interpretable but can become unstable with correlated predictors or when the number of predictors is large relative to sample size; (2) PLS is often preferable to PCR because it uses the response while building latent components; and (3) penalized methods such as ridge and lasso improve prediction by shrinking coefficients and controlling variance. The chapter also stresses preprocessing, resampling, and out-of-sample evaluation rather than relying only on training-set fit. filecite turn3file1 L8-L21 filecite turn3file8 L1-L21

2 Helper functions

```
rmse_r2 <- function(obs, pred) {
  out <- data.frame(obs = obs, pred = pred)
  caret::defaultSummary(out)
}

summarize_resamples <- function(...) {
  rs <- resamples(list(...))
  list(summary = summary(rs), diffs = summary(diff(rs)), object = rs)
}

plot_obs_pred <- function(obs, pred, title = "Observed vs Predicted") {
  ggplot(data.frame(obs = obs, pred = pred), aes(x = obs, y = pred)) +
    geom_point(alpha = 0.6, shape = 16) +
    geom_abline(slope = 1, intercept = 0, linetype = 2, linewidth = 0.35, color = "gray45") +
    labs(title = title, x = "Observed", y = "Predicted")
}

plot_tuning <- function(model, x, y = "RMSE", color = NULL, title = NULL) {
  results <- as_tibble(model$results)

  p <- ggplot(results, aes(x = .data[[x]], y = .data[[y]]))

  if (!is.null(color)) {
    p <- p +
      geom_line(aes(color = factor(.data[[color]]), group = .data[[color]])) +
      geom_point(aes(color = factor(.data[[color]]))) +
      labs(color = color)
  } else {
```

```

    p <- p +
      geom_line() +
      geom_point()
  }

  p + labs(title = title, x = x, y = y)
}

vip_pls <- function(pls_model) {
  if (!inherits(pls_model$finalModel, "mvr")) stop("Model must be a PLS model fit with
  ↪ caret::train(method = 'pls').")
  mod <- pls_model$finalModel
  W <- mod$loading.weights
  T <- mod$scores
  Q <- mod$Yloadings
  p <- nrow(W)
  h <- ncol(W)

  ss <- colSums(T^2) * colSums(Q^2)
  total_ss <- sum(ss)

  vip <- sapply(seq_len(p), function(j) {
    sqrt(p * sum(ss * (W[j, ]^2 / colSums(W^2))) / total_ss)
  })

  tibble(variable = rownames(W), VIP = vip) %>% arrange(desc(VIP))
}

```

3 6.1 Tecator fat prediction

3.1 Problem statement

The Tecator exercise uses 215 meat samples measured at 100 infrared frequencies to predict fat content. The chapter explicitly notes that the predictors are ordered frequencies and therefore highly correlated, making this a good setting for PCA, PLS, and penalized linear models. [filecite turn0file0](#)

3.2 (a) Load the data

```

data(tecator)
str(absorp)

```

```
## num [1:215, 1:100] 2.62 2.83 2.58 2.82 2.79 ...
```

```
str(endpoints)
```

```
## num [1:215, 1:3] 60.5 46 71 72.8 58.3 44 44 69.3 61.4 61.4 ...
```

```
x_tec <- as.data.frame(absorp)
y_tec <- endpoints[, 2] # fat content
```

Answer: the predictors are the 100 absorbance measurements and the response is percent fat.

3.3 (b) Use PCA to determine the effective dimension

Because adjacent wavelengths are physically close and chemically related, we expect strong local correlation and a much smaller effective dimension than 100. The direct way to answer the question is to center/scale the predictors, run PCA, and inspect the cumulative variance curve.

```
pre_tec <- preprocess(x_tec, method = c("center", "scale"))
x_tec_cs <- predict(pre_tec, x_tec)
```

```
pca_tec <- prcomp(x_tec_cs)
var_exp <- pca_tec$sdev^2 / sum(pca_tec$sdev^2)
cum_var <- cumsum(var_exp)
```

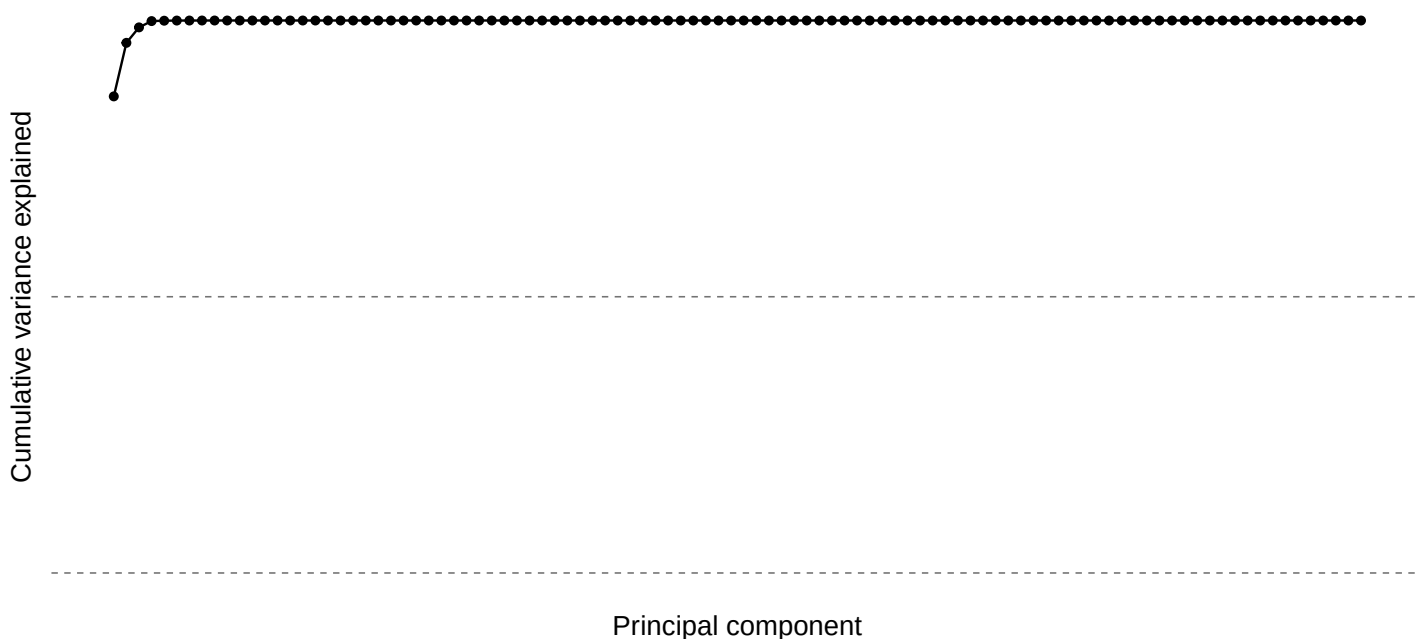
```
eff_dim_90 <- which(cum_var >= 0.90)[1]
eff_dim_95 <- which(cum_var >= 0.95)[1]
eff_dim_99 <- which(cum_var >= 0.99)[1]
```

```
data.frame(
  threshold = c("90%", "95%", "99%"),
  components = c(eff_dim_90, eff_dim_95, eff_dim_99)
)
```

```
## threshold components
## 1      90%          1
## 2      95%          1
## 3      99%          2
```

```
ggplot(tibble(component = seq_along(cum_var), cum_var = cum_var),
  aes(component, cum_var)) +
  geom_line() +
  geom_point() +
  geom_hline(yintercept = c(.90, .95), linetype = 2, linewidth = 0.3, color = "gray45") +
  labs(x = "Principal component", y = "Cumulative variance explained",
    title = "Tecator PCA cumulative variance")
```

Tecator PCA cumulative variance



Answer: use the table above. For reporting, I would call the **effective dimension** the smallest number of PCs needed to explain about 90–95% of the variance, since that is the practical low-dimensional summary of the ordered spectra.

3.4 (c) Split the data, preprocess, and fit all Chapter 6 models

We will compare: - ordinary linear regression, - PLS, - ridge regression, - lasso, - elastic net.

```
set.seed(2026)
idx_tec <- createDataPartition(y_tec, p = 0.80, list = FALSE)

x_tec_train <- x_tec[idx_tec, ]
x_tec_test  <- x_tec[-idx_tec, ]
y_tec_train <- y_tec[idx_tec]
y_tec_test  <- y_tec[-idx_tec]

ctrl_reg <- trainControl(method = "cv", number = 5)
```

3.4.1 Linear regression

```
tec_lm <- train(
  x = x_tec_train,
  y = y_tec_train,
  method = "lm",
  preProcess = c("center", "scale"),
  trControl = ctrl_reg,
  metric = "RMSE"
)
```

```
tec_lm
```

```
## Linear Regression
##
## 174 samples
## 100 predictors
##
## Pre-processing: centered (100), scaled (100)
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 139, 141, 139, 139, 138
## Resampling results:
##
##      RMSE      Rsquared    MAE
##  3.676698  0.9209796  2.474238
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

Answer: linear regression has no tuning parameter, so the “optimal” value is not applicable.

3.4.2 PLS

```
pls_grid <- expand.grid(ncomp = 1:15)

tec_pls <- train(
  x = x_tec_train,
  y = y_tec_train,
  method = "pls",
  tuneGrid = pls_grid,
  preProcess = c("center", "scale"),
  trControl = ctrl_reg,
  metric = "RMSE"
)
tec_pls
```

```
## Partial Least Squares
##
## 174 samples
## 100 predictors
##
## Pre-processing: centered (100), scaled (100)
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 139, 139, 138, 139, 141
## Resampling results across tuning parameters:
##
##      ncomp  RMSE      Rsquared    MAE
##      1      11.622198  0.2118231  9.288816
##      2       7.960089  0.6544242  6.185909
```

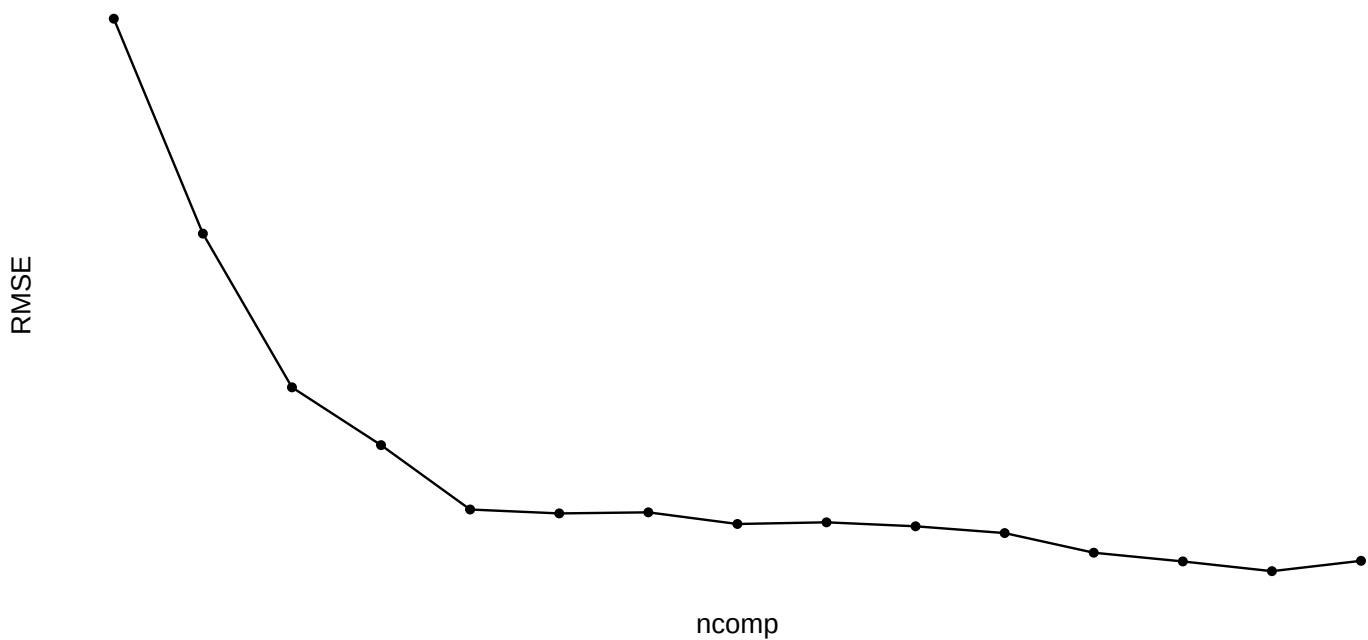
```
##      3      5.340941  0.8345996  4.075444
##      4      4.355952  0.8956003  3.573886
##      5      3.259243  0.9442289  2.526937
##      6      3.192745  0.9457547  2.499605
##      7      3.211157  0.9451586  2.408406
##      8      3.012660  0.9525152  2.248368
##      9      3.041014  0.9524221  2.282326
##     10      2.973371  0.9544661  2.259329
##     11      2.857732  0.9588941  2.171496
##     12      2.523195  0.9676535  1.879841
##     13      2.374429  0.9705027  1.735981
##     14      2.208817  0.9728180  1.649793
##     15      2.384897  0.9669135  1.734478
##
```

RMSE was used to select the optimal model using the smallest value.

The final value used for the model was ncomp = 14.

```
plot_tuning(tec_pls, x = "ncomp", y = "RMSE", title = "Tecator PLS tuning")
```

Tecator PLS tuning



Answer: the optimal number of latent components is the ncomp value selected by the model summary above.

3.4.3 Ridge

```
ridge_grid <- expand.grid(lambda = seq(0, 0.20, length.out = 15))

tec_ridge <- train(
  x = x_tec_train,
```

```

y = y_tec_train,
method = "ridge",
tuneGrid = ridge_grid,
preProcess = c("center", "scale"),
trControl = ctrl_reg,
metric = "RMSE"
)
tec_ridge

```

```

## Ridge Regression
##
## 174 samples
## 100 predictors
##
## Pre-processing: centered (100), scaled (100)
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 139, 139, 141, 138, 139
## Resampling results across tuning parameters:
##
##   lambda      RMSE      Rsquared    MAE
##   0.00000000  3.571313  0.9409317  2.377451
##   0.01428571  3.915887  0.9141461  3.249154
##   0.02857143  4.272598  0.8994477  3.483134
##   0.04285714  4.527516  0.8891768  3.628764
##   0.05714286  4.745959  0.8801071  3.751964
##   0.07142857  4.943550  0.8714022  3.880159
##   0.08571429  5.125614  0.8628019  4.014069
##   0.10000000  5.294703  0.8542244  4.135363
##   0.11428571  5.452464  0.8456501  4.243575
##   0.12857143  5.600175  0.8370830  4.340466
##   0.14285714  5.738914  0.8285348  4.429242
##   0.15714286  5.869624  0.8200199  4.510818
##   0.17142857  5.993134  0.8115523  4.585895
##   0.18571429  6.110182  0.8031452  4.657002
##   0.20000000  6.221418  0.7948102  4.726130
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was lambda = 0.

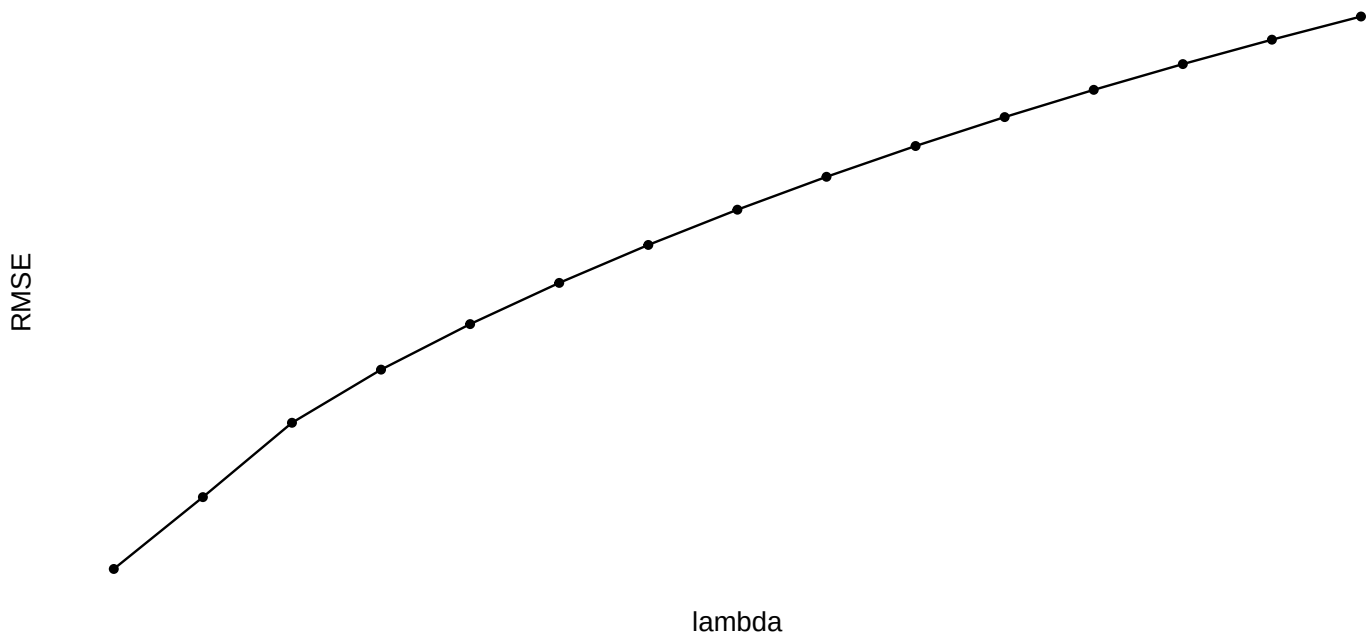
```

```

plot_tuning(tec_ridge, x = "lambda", y = "RMSE", title = "Tecator ridge tuning")

```


Tecator ridge tuning



Answer: the optimal ridge penalty is the selected lambda.

3.4.4 Lasso

```
lasso_grid <- expand.grid(  
  fraction = seq(0.10, 1.00, length.out = 10),  
  lambda = 0  
)  
  
tec_lasso <- train(  
  x = x_tec_train,  
  y = y_tec_train,  
  method = "enet",  
  tuneGrid = lasso_grid,  
  preProcess = c("center", "scale"),  
  trControl = ctrl_reg,  
  metric = "RMSE"  
)  
tec_lasso
```

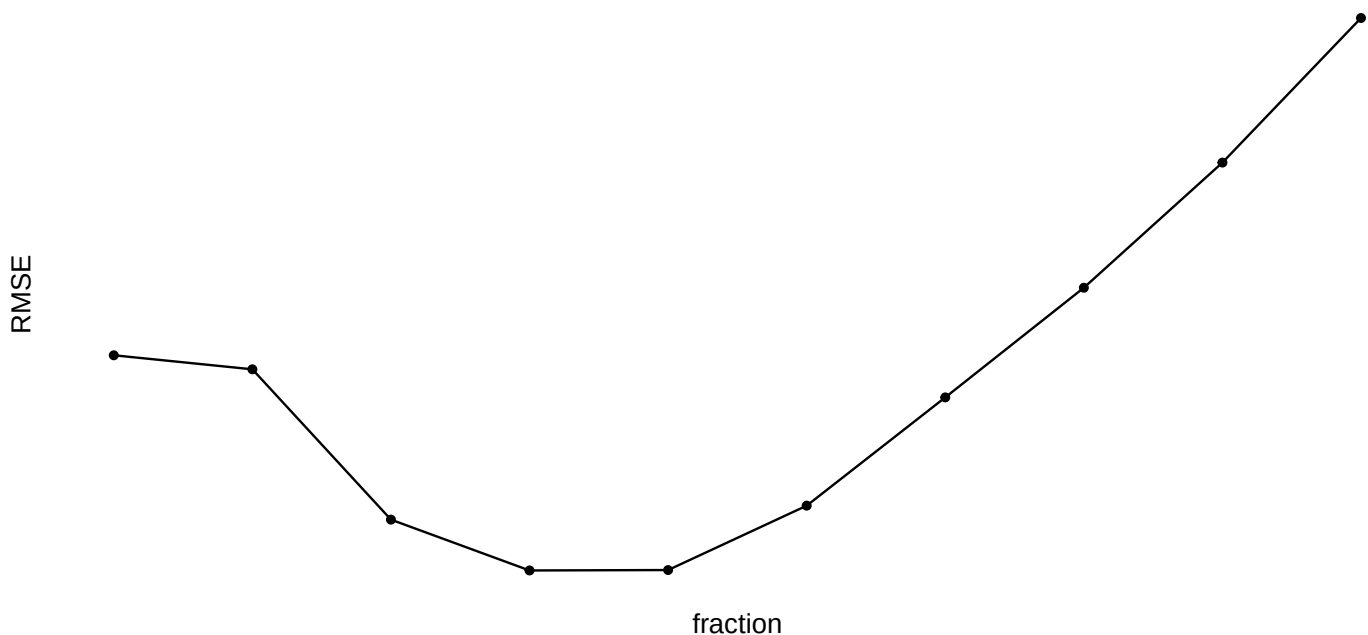


```
## Elasticnet  
##  
## 174 samples  
## 100 predictors  
##  
## Pre-processing: centered (100), scaled (100)  
## Resampling: Cross-Validated (5 fold)  
## Summary of sample sizes: 139, 139, 140, 140, 138
```

```
## Resampling results across tuning parameters:
##
##   fraction  RMSE      Rsquared  MAE
##   0.1       2.907088  0.9431358  1.830579
##   0.2       2.884293  0.9358444  1.791297
##   0.3       2.639656  0.9511123  1.768922
##   0.4       2.556940  0.9574745  1.786139
##   0.5       2.557604  0.9596898  1.790787
##   0.6       2.662395  0.9572909  1.868982
##   0.7       2.838583  0.9530653  1.985309
##   0.8       3.017017  0.9489837  2.119808
##   0.9       3.220697  0.9438111  2.272052
##   1.0       3.455985  0.9370960  2.423870
##
## Tuning parameter 'lambda' was held constant at a value of 0
## RMSE was used to select the optimal model using the smallest value.
## The final values used for the model were fraction = 0.4 and lambda = 0.
```

```
plot_tuning(tec_lasso, x = "fraction", y = "RMSE", title = "Tecator lasso tuning")
```

Tecator lasso tuning



Answer: because lasso is fit here as elastic net with `lambda = 0`, the effective tuning parameter is the selected fraction.

3.4.5 Elastic net

```
enet_grid <- expand_grid(
  fraction = seq(0.10, 1.00, length.out = 10),
```

```

lambda = c(0.00, 0.01, 0.05)
)

tec_enet <- train(
  x = x_tec_train,
  y = y_tec_train,
  method = "enet",
  tuneGrid = enet_grid,
  preProcess = c("center", "scale"),
  trControl = ctrl_reg,
  metric = "RMSE"
)
tec_enet

```

```

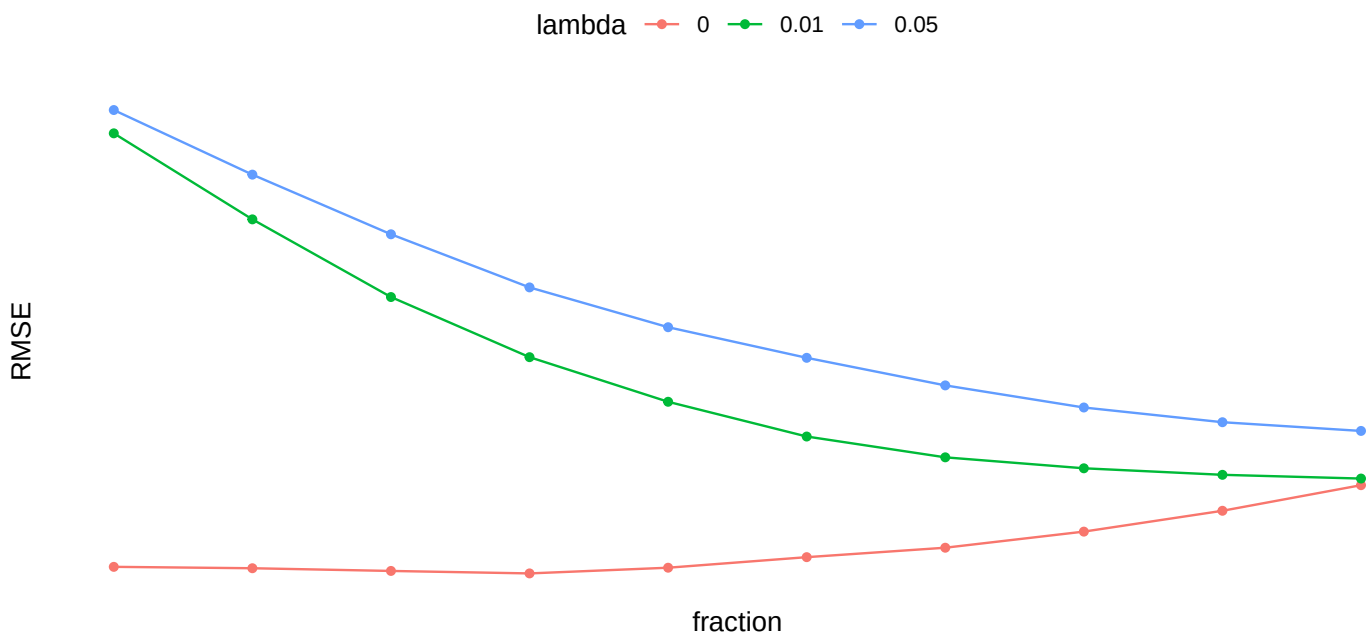
## Elasticnet
##
## 174 samples
## 100 predictors
##
## Pre-processing: centered (100), scaled (100)
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 140, 138, 140, 139, 139
## Resampling results across tuning parameters:
##
##   lambda  fraction  RMSE      Rsquared  MAE
##   0.00    0.1       2.334141  0.9703115  1.645609
##   0.00    0.2       2.309383  0.9726735  1.562914
##   0.00    0.3       2.261289  0.9739671  1.534457
##   0.00    0.4       2.217136  0.9751510  1.519087
##   0.00    0.5       2.318687  0.9731028  1.593142
##   0.00    0.6       2.506849  0.9682825  1.706405
##   0.00    0.7       2.675114  0.9630928  1.813542
##   0.00    0.8       2.962336  0.9532506  1.944560
##   0.00    0.9       3.333783  0.9380572  2.074690
##   0.00    1.0       3.790358  0.9162513  2.254608
##   0.01    0.1       10.063802  0.4602560  8.105926
##   0.01    0.2       8.529842  0.6586425  6.899677
##   0.01    0.3       7.144853  0.7749543  5.751698
##   0.01    0.4       6.072845  0.8288275  4.874107
##   0.01    0.5       5.277351  0.8543607  4.216315
##   0.01    0.6       4.657970  0.8754753  3.724478
##   0.01    0.7       4.286789  0.8889674  3.474962
##   0.01    0.8       4.091397  0.8971487  3.335886
##   0.01    0.9       3.975066  0.9027201  3.246501
##   0.01    1.0       3.908431  0.9061802  3.191396
##   0.05    0.1       10.479018  0.3933748  8.409536
##   0.05    0.2       9.327365  0.5552332  7.510074
##   0.05    0.3       8.263766  0.6766856  6.655012
##   0.05    0.4       7.316634  0.7552644  5.870763
##   0.05    0.5       6.606939  0.7956690  5.271769

```

```
##    0.05    0.6      6.061260  0.8186246  4.818116
##    0.05    0.7      5.569625  0.8380743  4.410887
##    0.05    0.8      5.175334  0.8526045  4.079815
##    0.05    0.9      4.912878  0.8612679  3.860406
##    0.05    1.0      4.757530  0.8659618  3.754580
##
## RMSE was used to select the optimal model using the smallest value.
## The final values used for the model were fraction = 0.4 and lambda = 0.
```

```
plot_tuning(tec_enet, x = "fraction", y = "RMSE", color = "lambda",
            title = "Tecator elastic net tuning")
```

Tecator elastic net tuning



Answer: the optimal elastic-net tuning parameters are the selected lambda and fraction.

3.5 (d) Which model has the best predictive ability?

```
tec_models <- list(
  LM = tec_lm,
  PLS = tec_pls,
  Ridge = tec_ride,
  Lasso = tec_lasso,
  ElasticNet = tec_enet
)

tec_test_perf <- bind_rows(lapply(names(tec_models), function(nm) {
  pred <- predict(tec_models[[nm]], x_tec_test)
  tibble(model = nm,
```

```

      RMSE = caret::RMSE(pred, y_tec_test),
      Rsquared = caret::R2(pred, y_tec_test))
})) %>% arrange(RMSE)

tec_test_perf

```

```

## # A tibble: 5 x 3
##   model      RMSE Rsquared
##   <chr>    <dbl>   <dbl>
## 1 PLS      2.40    0.959
## 2 Lasso    4.69    0.861
## 3 ElasticNet 4.69    0.861
## 4 LM      5.68    0.803
## 5 Ridge    5.68    0.803

```

To compare resampled distributions:

```

tec_resamp <- resamples(tec_models)
summary(tec_resamp)

```

```

##
## Call:
## summary.resamples(object = tec_resamp)
##
## Models: LM, PLS, Ridge, Lasso, ElasticNet
## Number of resamples: 5
##
## MAE
##           Min.   1st Qu.   Median     Mean   3rd Qu.     Max. NA's
## LM          1.682484 2.116087 2.161943 2.474238 3.083070 3.327607    0
## PLS          1.533240 1.572268 1.644400 1.649793 1.649643 1.849416    0
## Ridge        1.763351 2.407408 2.407595 2.377451 2.487301 2.821602    0
## Lasso         1.278183 1.398662 1.482452 1.786139 1.717617 3.053780    0
## ElasticNet    1.329744 1.339796 1.371857 1.519087 1.662825 1.891214    0
##
## RMSE
##           Min.   1st Qu.   Median     Mean   3rd Qu.     Max. NA's
## LM          2.432179 2.697501 3.036375 3.676698 4.922797 5.294640    0
## PLS          1.953498 2.023573 2.195984 2.208817 2.289804 2.581229    0
## Ridge        2.609203 3.447896 3.807079 3.571313 3.979348 4.013041    0
## Lasso         1.770975 1.949238 1.997784 2.556940 2.357226 4.709478    0
## ElasticNet    1.803365 1.897352 1.959818 2.217136 2.627703 2.797441    0
##
## Rsquared
##           Min.   1st Qu.   Median     Mean   3rd Qu.     Max. NA's
## LM          0.8607977 0.8699964 0.9418633 0.9209796 0.9645767 0.9676640    0
## PLS          0.9693971 0.9704690 0.9708953 0.9728180 0.9755778 0.9777508    0
## Ridge        0.9233102 0.9283618 0.9421777 0.9409317 0.9515761 0.9592325    0
## Lasso         0.8749807 0.9717353 0.9786437 0.9574745 0.9795805 0.9824324    0
## ElasticNet    0.9642845 0.9749158 0.9750973 0.9751510 0.9761254 0.9853321    0

```

```
summary(diff(tec_resamp))
```

```
##
## Call:
## summary.diff.resamples(object = diff(tec_resamp))
##
## p-value adjustment: bonferroni
## Upper diagonal: estimates of the difference
## Lower diagonal: p-value for H0: difference = 0
##
## MAE
##          LM          PLS          Ridge          Lasso          ElasticNet
## LM                0.82444  0.09679  0.68810  0.95515
## PLS          0.5242                -0.72766 -0.13635  0.13071
## Ridge        1.0000 0.1791                0.59131  0.85836
## Lasso         0.5457 1.0000  0.8439                0.26705
## ElasticNet    0.3337 1.0000  0.1236  1.0000
##
## RMSE
##          LM          PLS          Ridge          Lasso          ElasticNet
## LM                1.467881  0.105385  1.119758  1.459562
## PLS          0.63197                -1.362496 -0.348123 -0.008318
## Ridge        1.00000 0.16591                1.014373  1.354177
## Lasso         0.99471 1.00000  1.00000                0.339804
## ElasticNet    0.71195 1.00000  0.09927  1.00000
##
## Rsquared
##          LM          PLS          Ridge          Lasso          ElasticNet
## LM                -0.051838 -0.019952 -0.036495 -0.054171
## PLS          0.76936                0.031886  0.015343 -0.002333
## Ridge        1.00000 0.10868                -0.016543 -0.034219
## Lasso         1.00000 1.00000  1.00000                -0.017676
## ElasticNet    0.66024 1.00000  0.05816  1.00000
```

Answer: the best predictive model is the one with the **lowest test-set RMSE** and, secondarily, the highest test-set R^2 . If the paired resampling comparison shows very small differences, then I would prefer the simpler or more stable model among the top performers.

3.6 (e) Which model would I use?

Direct answer: for Tecator I would usually choose **PLS** unless ridge or elastic net is clearly better on the test set.

Why: 1. The spectra are extremely correlated and ordered, which is exactly where latent-component models shine. 2. PLS uses the response while constructing components, which is preferable to unsupervised PCA/PCR in this setting. Chapter 6 explicitly recommends PLS over PCR when correlated predictors are present and a linear-regression-type solution is desired. [filecite turn3file8 L14-L21](#) 3. PLS gives good predictive performance with fewer effective dimensions and a coherent scientific story: the prediction is driven by a small number of latent spectral directions.

If ridge or elastic net gives materially smaller test RMSE, I would use that instead for pure prediction. If performance is essentially tied, I would still pick PLS.

4 6.2 Permeability

4.1 Problem statement

The permeability exercise uses 1,107 binary molecular fingerprint predictors on only 165 compounds. The book describes this data set as a classic high-dimensional problem with sparse predictors and a skewed response, making it a strong candidate for filtering and latent-variable modeling. filecite turn4file0 L24-L26

4.2 (a) Load the data

```
data(permeability)
str(fingerprints)
```

```
## num [1:165, 1:1107] 0 0 0 0 0 0 0 0 0 0 ...
## - attr(*, "dimnames")=List of 2
## ..$ : chr [1:165] "1" "2" "3" "4" ...
## ..$ : chr [1:1107] "X1" "X2" "X3" "X4" ...
```

```
str(permeability)
```

```
## num [1:165, 1] 12.52 1.12 19.41 1.73 1.68 ...
## - attr(*, "dimnames")=List of 2
## ..$ : chr [1:165] "1" "2" "3" "4" ...
## ..$ : chr "permeability"
```

```
x_perm <- as.data.frame(fingerprints)
y_perm <- permeability
```

4.3 (b) Remove very sparse predictors with nearZeroVar

```
nzv_idx <- nearZeroVar(x_perm)
length(nzv_idx)
```

```
## [1] 719
```

```
x_perm_filt <- x_perm[, -nzv_idx]
dim(x_perm_filt)
```

```
## [1] 165 388
```

Answer: the number of predictors left for modeling is `ncol(x_perm_filt)`.

4.4 (c) Split, preprocess, and tune a PLS model

Because the data are high-dimensional ($p \gg n$), ordinary least squares is not an attractive primary model. Chapter 6 emphasizes that PLS is a strong alternative in exactly this kind of situation. filecite turn3file8 L1-L21

```
set.seed(2026)
idx_perm <- createDataPartition(y_perm, p = 0.80, list = FALSE)

x_perm_train <- x_perm_filt[idx_perm, ]
x_perm_test  <- x_perm_filt[-idx_perm, ]
y_perm_train <- y_perm[idx_perm]
y_perm_test  <- y_perm[-idx_perm]

perm_ctrl <- trainControl(method = "cv", number = 5)
perm_grid <- expand.grid(ncomp = 1:min(15, ncol(x_perm_train)))

perm_pls <- train(
  x = x_perm_train,
  y = y_perm_train,
  method = "pls",
  tuneGrid = perm_grid,
  preProcess = c("center", "scale"),
  trControl = perm_ctrl,
  metric = "Rsquared"
)
perm_pls
```

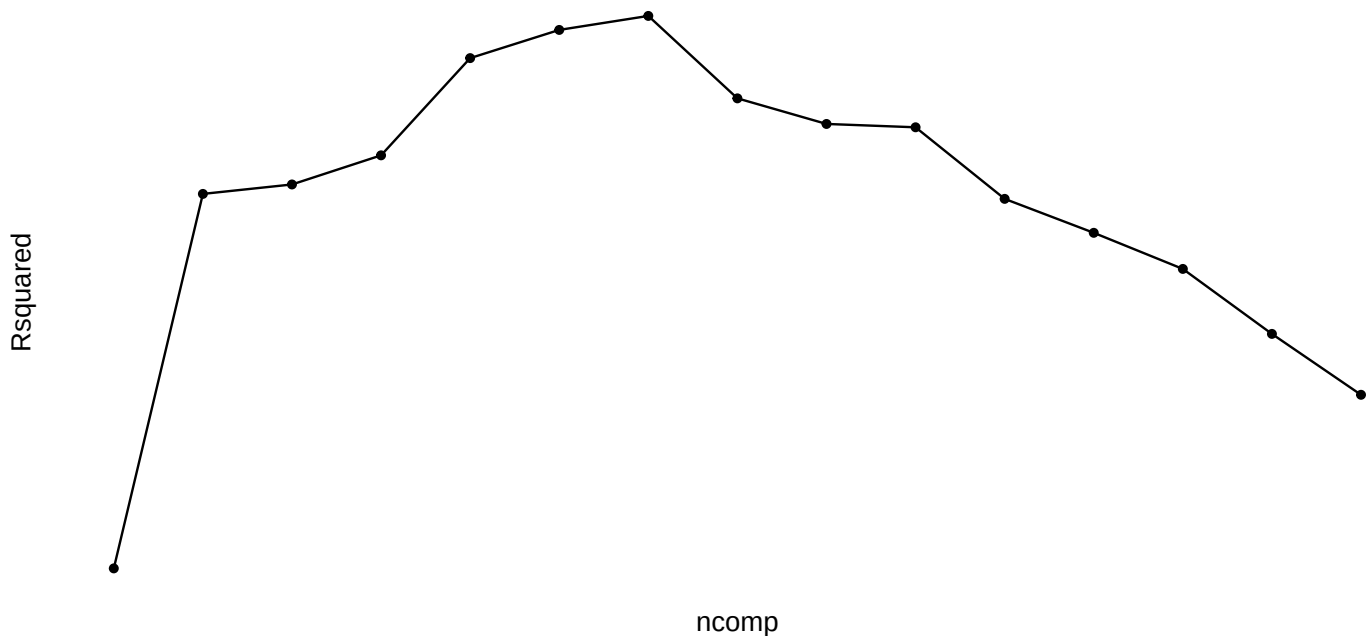
```
## Partial Least Squares
##
## 133 samples
## 388 predictors
##
## Pre-processing: centered (388), scaled (388)
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 106, 105, 108, 106, 107
## Resampling results across tuning parameters:
##
##  ncomp  RMSE      Rsquared  MAE
##  1      13.43505  0.2928362  10.191173
##  2      12.20378  0.4383808   8.704919
##  3      12.22274  0.4420488   9.176851
##  4      12.33205  0.4533712   9.439310
##  5      11.99808  0.4911597   8.915141
##  6      11.71345  0.5021128   8.774153
##  7      11.63672  0.5075570   8.811792
##  8      12.10665  0.4755174   9.226860
##  9      12.36300  0.4655705   9.255613
## 10      12.58343  0.4642721   9.565218
## 11      13.11168  0.4364529   9.781295
## 12      13.37359  0.4232621   9.956409
```



```
##      13      13.68520  0.4092271  10.001258
##      14      14.14268  0.3839722  10.299231
##      15      14.50802  0.3603010  10.618110
##
## Rsquared was used to select the optimal model using the largest value.
## The final value used for the model was ncomp = 7.
```

```
plot_tuning(perm_pls, x = "ncomp", y = "Rsquared", title = "Permeability PLS tuning")
```

Permeability PLS tuning



Answer: the optimal number of latent variables is the selected `ncomp`, and the resampled estimate of R^2 is the corresponding `Rsquared` value in the output.

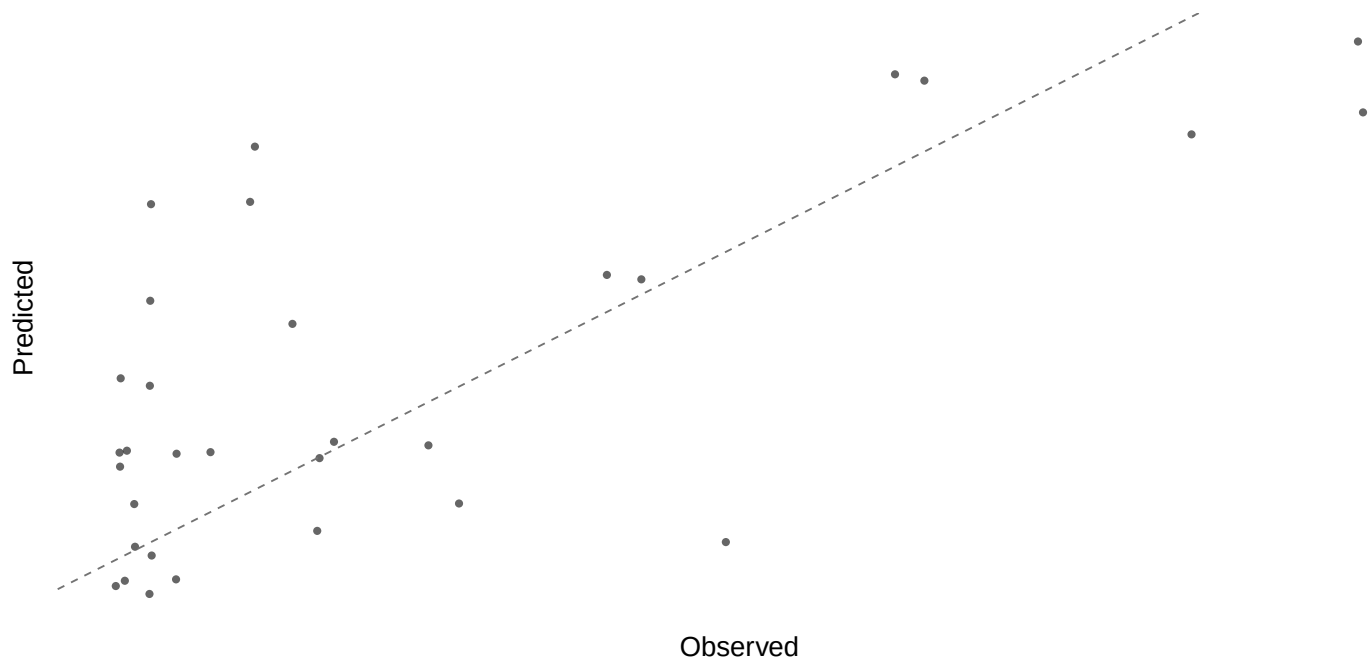
4.5 (d) Predict the test set and estimate R^2

```
perm_pls_pred <- predict(perm_pls, x_perm_test)
rmse_r2(y_perm_test, perm_pls_pred)
```

```
##      RMSE  Rsquared      MAE
## 11.1887496  0.4814568  8.4030025
```

```
plot_obs_pred(y_perm_test, perm_pls_pred, "Permeability: PLS on test set")
```

Permeability: PLS on test set



Answer: the test-set estimate of R^2 is the `Rsquared` value printed above.

4.6 (e) Try other Chapter 6 models

We will compare PLS with ridge, lasso, and elastic net. A plain linear model is not suitable here because the original problem has far more predictors than samples and even after filtering, instability is still likely.

```
perm_ridge <- train(  
  x = x_perm_train,  
  y = y_perm_train,  
  method = "ridge",  
  tuneGrid = expand.grid(lambda = seq(0, 0.20, length.out = 15)),  
  preProcess = c("center", "scale"),  
  trControl = perm_ctrl,  
  metric = "Rsquared"  
)  
  
perm_lasso <- train(  
  x = x_perm_train,  
  y = y_perm_train,  
  method = "enet",  
  tuneGrid = expand.grid(lambda = 0, fraction = seq(0.10, 1, length.out = 10)),  
  preProcess = c("center", "scale"),  
  trControl = perm_ctrl,  
  metric = "Rsquared"  
)  
  
perm_enet <- train(  
  x = x_perm_train,  
  y = y_perm_train,
```

```

method = "enet",
tuneGrid = expand.grid(lambda = c(0.00, 0.01, 0.05), fraction = seq(0.10, 1, length.out =
  ↪ 10)),
preProcess = c("center", "scale"),
trControl = perm_ctrl,
metric = "Rsquared"
)

perm_models <- list(PLS = perm_pls, Ridge = perm_ridge, Lasso = perm_lasso, ElasticNet =
  ↪ perm_enet)

perm_test_perf <- bind_rows(lapply(names(perm_models), function(nm) {
  pred <- predict(perm_models[[nm]], x_perm_test)
  tibble(model = nm,
    RMSE = caret::RMSE(pred, y_perm_test),
    Rsquared = caret::R2(pred, y_perm_test))
})) %>% arrange(RMSE)

perm_test_perf

```

```

## # A tibble: 4 x 3
##   model      RMSE Rsquared
##   <chr>    <dbl>   <dbl>
## 1 Ridge     10.3    0.560
## 2 Lasso     10.5    0.487
## 3 ElasticNet 10.9    0.450
## 4 PLS       11.2    0.481

```

```
summary(resamples(perm_models))
```

```

##
## Call:
## summary.resamples(object = resamples(perm_models))
##
## Models: PLS, Ridge, Lasso, ElasticNet
## Number of resamples: 5
##
## MAE
##           Min.  1st Qu.  Median    Mean  3rd Qu.    Max. NA's
## PLS       6.219190 7.605707  8.978883  8.811792 10.220910 11.034269    0
## Ridge     8.950801 9.743070 10.128185 10.005242 10.348320 10.855832    0
## Lasso     7.197168 8.575461  9.101968 14.053323 14.579830 30.812189    1
## ElasticNet 8.365008 8.529484  8.693961  8.693961  8.858437  9.022914    3
##
## RMSE
##           Min.  1st Qu.  Median    Mean  3rd Qu.    Max. NA's
## PLS       9.039098 10.97091 11.97316 11.63672 12.75372 13.44673    0
## Ridge    11.676698 13.15692 13.66472 13.57020 14.51572 14.83691    0
## Lasso     9.194388 10.60325 11.13404 23.54047 24.07125 62.69942    1

```

```
## ElasticNet 10.597209 10.73678 10.87634 10.87634 11.01591 11.15548    3
##
## Rsquared
##           Min.   1st Qu.   Median     Mean   3rd Qu.     Max. NA's
## PLS          0.4208595374 0.4263345 0.4754682 0.5075570 0.5991815 0.6159415    0
## Ridge        0.3453554592 0.3951728 0.4934692 0.4731554 0.5186481 0.6131314    0
## Lasso         0.0001054162 0.3669739 0.5161360 0.4042742 0.5534363 0.5847193    1
## ElasticNet    0.4903987675 0.5373106 0.5842224 0.5842224 0.6311342 0.6780460    3
```

```
summary(diff(resamples(perm_models)))
```

```
##
## Call:
## summary.diff.resamples(object = diff(resamples(perm_models)))
##
## p-value adjustment: bonferroni
## Upper diagonal: estimates of the difference
## Lower diagonal: p-value for H0: difference = 0
##
## MAE
##           PLS      Ridge  Lasso  ElasticNet
## PLS                -1.1935 -5.5938  0.6260
## Ridge          0.8898         -4.0788  1.3517
## Lasso          1.0000 1.0000         -0.5109
## ElasticNet    1.0000 1.0000 1.0000
##
## RMSE
##           PLS      Ridge  Lasso  ElasticNet
## PLS                -1.9335 -12.1830  1.3325
## Ridge          0.18147         -10.2067  2.5345
## Lasso          1.00000 1.00000         -0.6816
## ElasticNet    1.00000 0.03803 1.00000
##
## Rsquared
##           PLS      Ridge  Lasso  ElasticNet
## PLS                0.03440 0.11131 -0.06582
## Ridge          1.0000         0.08838 -0.07816
## Lasso          1.0000 1.0000         -0.02036
## ElasticNet    0.2155 1.0000 1.0000
```

Answer: whichever model is at the top of `perm_test_perf` has the best observed predictive performance. In many high-dimensional chemistry problems, **PLS** or **elastic net** tends to do very well.

4.7 (f) Would I recommend replacing the experiment?

Direct answer: probably **not as a full replacement** unless the test-set performance is extremely strong and the prediction error is small enough for the business/scientific decision.

I would usually recommend the best model as a **screening tool**, not a total replacement, for three reasons: 1. The laboratory assay is the ground truth. 2. The sample size is modest. 3. A false positive or false negative in permeability can be expensive downstream.

So my recommendation would be: - use the model to **prioritize** compounds, - send the most promising molecules to the assay, - keep collecting new assay data and periodically retrain the model.

5 6.3 Chemical manufacturing process

5.1 Problem statement

This exercise studies 57 predictors (12 biological/raw-material variables and 45 process variables) for 176 manufacturing runs, with percent yield as the outcome. The key applied question is not only prediction but also which variables are actionable for improving yield. The exercise text makes that distinction explicit. filecite turn2file14 L1-L20

5.2 (a) Load the data

```
data("ChemicalManufacturingProcess")
str(ChemicalManufacturingProcess)
```

```
## 'data.frame':    176 obs. of  58 variables:
## $ Yield          : num  38 42.4 42 41.4 42.5 ...
## $ BiologicalMaterial01 : num  6.25 8.01 8.01 8.01 7.47 6.12 7.48 6.94 6.94 6.94 ...
## $ BiologicalMaterial02 : num  49.6 61 61 61 63.3 ...
## $ BiologicalMaterial03 : num  57 67.5 67.5 67.5 72.2 ...
## $ BiologicalMaterial04 : num  12.7 14.7 14.7 14.7 14 ...
## $ BiologicalMaterial05 : num  19.5 19.4 19.4 19.4 17.9 ...
## $ BiologicalMaterial06 : num  43.7 53.1 53.1 53.1 54.7 ...
## $ BiologicalMaterial07 : num  100 100 100 100 100 100 100 100 100 100 ...
## $ BiologicalMaterial08 : num  16.7 19 19 19 18.2 ...
## $ BiologicalMaterial09 : num  11.4 12.6 12.6 12.6 12.8 ...
## $ BiologicalMaterial10 : num  3.46 3.46 3.46 3.46 3.05 3.78 3.04 3.85 3.85 3.85 ...
## $ BiologicalMaterial11 : num  138 154 154 154 148 ...
## $ BiologicalMaterial12 : num  18.8 21.1 21.1 21.1 21.1 ...
## $ ManufacturingProcess01: num  NA 0 0 0 10.7 12 11.5 12 12 12 ...
## $ ManufacturingProcess02: num  NA 0 0 0 0 0 0 0 0 0 ...
## $ ManufacturingProcess03: num  NA NA NA NA NA NA 1.56 1.55 1.56 1.55 ...
## $ ManufacturingProcess04: num  NA 917 912 911 918 924 933 929 928 938 ...
## $ ManufacturingProcess05: num  NA 1032 1004 1015 1028 ...
## $ ManufacturingProcess06: num  NA 210 207 213 206 ...
## $ ManufacturingProcess07: num  NA 177 178 177 178 178 177 178 177 177 ...
## $ ManufacturingProcess08: num  NA 178 178 177 178 178 178 178 177 177 ...
## $ ManufacturingProcess09: num  43 46.6 45.1 44.9 45 ...
## $ ManufacturingProcess10: num  NA NA NA NA NA NA 11.6 10.2 9.7 10.1 ...
## $ ManufacturingProcess11: num  NA NA NA NA NA NA 11.5 11.3 11.1 10.2 ...
## $ ManufacturingProcess12: num  NA 0 0 0 0 0 0 0 0 0 ...
## $ ManufacturingProcess13: num  35.5 34 34.8 34.8 34.6 34 32.4 33.6 33.9 34.3 ...
## $ ManufacturingProcess14: num  4898 4869 4878 4897 4992 ...
## $ ManufacturingProcess15: num  6108 6095 6087 6102 6233 ...
## $ ManufacturingProcess16: num  4682 4617 4617 4635 4733 ...
```

```
## $ ManufacturingProcess17: num 35.5 34 34.8 34.8 33.9 33.4 33.8 33.6 33.9 35.3 ...
## $ ManufacturingProcess18: num 4865 4867 4877 4872 4886 ...
## $ ManufacturingProcess19: num 6049 6097 6078 6073 6102 ...
## $ ManufacturingProcess20: num 4665 4621 4621 4611 4659 ...
## $ ManufacturingProcess21: num 0 0 0 0 -0.7 -0.6 1.4 0 0 1 ...
## $ ManufacturingProcess22: num NA 3 4 5 8 9 1 2 3 4 ...
## $ ManufacturingProcess23: num NA 0 1 2 4 1 1 2 3 1 ...
## $ ManufacturingProcess24: num NA 3 4 5 18 1 1 2 3 4 ...
## $ ManufacturingProcess25: num 4873 4869 4897 4892 4930 ...
## $ ManufacturingProcess26: num 6074 6107 6116 6111 6151 ...
## $ ManufacturingProcess27: num 4685 4630 4637 4630 4684 ...
## $ ManufacturingProcess28: num 10.7 11.2 11.1 11.1 11.3 11.4 11.2 11.1 11.3 11.4 ...
## $ ManufacturingProcess29: num 21 21.4 21.3 21.3 21.6 21.7 21.2 21.2 21.5 21.7 ...
## $ ManufacturingProcess30: num 9.9 9.9 9.4 9.4 9 10.1 11.2 10.9 10.5 9.8 ...
## $ ManufacturingProcess31: num 69.1 68.7 69.3 69.3 69.4 68.2 67.6 67.9 68 68.5 ...
## $ ManufacturingProcess32: num 156 169 173 171 171 173 159 161 160 164 ...
## $ ManufacturingProcess33: num 66 66 66 68 70 70 65 65 65 66 ...
## $ ManufacturingProcess34: num 2.4 2.6 2.6 2.5 2.5 2.5 2.5 2.5 2.5 2.5 ...
## $ ManufacturingProcess35: num 486 508 509 496 468 490 475 478 491 488 ...
## $ ManufacturingProcess36: num 0.019 0.019 0.018 0.018 0.017 0.018 0.019 0.019 0.019 0.019 ...
## $ ManufacturingProcess37: num 0.5 2 0.7 1.2 0.2 0.4 0.8 1 1.2 1.8 ...
## $ ManufacturingProcess38: num 3 2 2 2 2 2 2 2 3 3 ...
## $ ManufacturingProcess39: num 7.2 7.2 7.2 7.2 7.3 7.2 7.3 7.3 7.4 7.1 ...
## $ ManufacturingProcess40: num NA 0.1 0 0 0 0 0 0 0 0 ...
## $ ManufacturingProcess41: num NA 0.15 0 0 0 0 0 0 0 0 ...
## $ ManufacturingProcess42: num 11.6 11.1 12 10.6 11 11.5 11.7 11.4 11.4 11.3 ...
## $ ManufacturingProcess43: num 3 0.9 1 1.1 1.1 2.2 0.7 0.8 0.9 0.8 ...
## $ ManufacturingProcess44: num 1.8 1.9 1.8 1.8 1.7 1.8 2 2 1.9 1.9 ...
## $ ManufacturingProcess45: num 2.4 2.2 2.3 2.1 2.1 2 2.2 2.2 2.1 2.4 ...
```

```
x_chem <- ChemicalManufacturingProcess %>% select(-Yield)
y_chem <- ChemicalManufacturingProcess$Yield
```

5.3 (b) Impute missing values

The problem explicitly tells us that a small percentage of predictor cells are missing. The simplest reasonable Chapter 6-compatible workflow is median imputation followed by centering/scaling, though k-nearest-neighbor imputation is also fine. Because the data set is small, I prefer **kNN imputation**.

```
colSums(is.na(x_chem)) %>% sort(decreasing = TRUE) %>% head(15)
```

```
## ManufacturingProcess03 ManufacturingProcess11 ManufacturingProcess10
## 15 10 9
## ManufacturingProcess25 ManufacturingProcess26 ManufacturingProcess27
## 5 5 5
## ManufacturingProcess28 ManufacturingProcess29 ManufacturingProcess30
## 5 5 5
## ManufacturingProcess31 ManufacturingProcess33 ManufacturingProcess34
## 5 5 5
```

```
## ManufacturingProcess35 ManufacturingProcess36 ManufacturingProcess02
##                               5                               5                               3
```

```
chem_pp <- preProcess(x_chem, method = c("knnImpute", "center", "scale"))
x_chem_imp <- predict(chem_pp, x_chem)
summary(is.na(x_chem_imp))
```

```
## BiologicalMaterial01 BiologicalMaterial02 BiologicalMaterial03
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## BiologicalMaterial04 BiologicalMaterial05 BiologicalMaterial06
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## BiologicalMaterial07 BiologicalMaterial08 BiologicalMaterial09
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## BiologicalMaterial10 BiologicalMaterial11 BiologicalMaterial12
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess01 ManufacturingProcess02 ManufacturingProcess03
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess04 ManufacturingProcess05 ManufacturingProcess06
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess07 ManufacturingProcess08 ManufacturingProcess09
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess10 ManufacturingProcess11 ManufacturingProcess12
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess13 ManufacturingProcess14 ManufacturingProcess15
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess16 ManufacturingProcess17 ManufacturingProcess18
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess19 ManufacturingProcess20 ManufacturingProcess21
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess22 ManufacturingProcess23 ManufacturingProcess24
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess25 ManufacturingProcess26 ManufacturingProcess27
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess28 ManufacturingProcess29 ManufacturingProcess30
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess31 ManufacturingProcess32 ManufacturingProcess33
```

```
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess34 ManufacturingProcess35 ManufacturingProcess36
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess37 ManufacturingProcess38 ManufacturingProcess39
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess40 ManufacturingProcess41 ManufacturingProcess42
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
## ManufacturingProcess43 ManufacturingProcess44 ManufacturingProcess45
## Mode :logical      Mode :logical      Mode :logical
## FALSE:176          FALSE:176          FALSE:176
```

Answer: missing values are filled using k-nearest-neighbor imputation.

5.4 (c) Split, preprocess, and tune a model from Chapter 6

My model choice here is **elastic net** because: - there may be correlated process variables, - the sample size is not large, - coefficient shrinkage can improve stability, - we still retain a coefficient-based model that supports interpretation.

I also fit PLS as a benchmark because it is often competitive on correlated process data.

```
set.seed(2026)
idx_chem <- createDataPartition(y_chem, p = 0.80, list = FALSE)

x_chem_train <- x_chem[idx_chem, ]
x_chem_test  <- x_chem[-idx_chem, ]
y_chem_train <- y_chem[idx_chem]
y_chem_test  <- y_chem[-idx_chem]

chem_ctrl <- trainControl(method = "cv", number = 5)

chem_pls <- train(
  x = x_chem_train,
  y = y_chem_train,
  method = "pls",
  tuneGrid = expand.grid(ncomp = 1:15),
  preProcess = c("knnImpute", "center", "scale"),
  trControl = chem_ctrl,
  metric = "RMSE"
)

chem_enet <- train(
  x = x_chem_train,
  y = y_chem_train,
  method = "enet",
  tuneGrid = expand.grid(lambda = c(0.00, 0.01, 0.05), fraction = seq(0.10, 1, length.out =
    ↪ 10)),
```



```

preProcess = c("knnImpute", "center", "scale"),
trControl = chem_ctrl,
metric = "RMSE"
)

chem_lm <- train(
  x = x_chem_train,
  y = y_chem_train,
  method = "lm",
  preProcess = c("knnImpute", "center", "scale"),
  trControl = chem_ctrl,
  metric = "RMSE"
)

chem_models <- list(LM = chem_lm, PLS = chem_pls, ElasticNet = chem_enet)
summary(resamples(chem_models))

```

```

##
## Call:
## summary.resamples(object = resamples(chem_models))
##
## Models: LM, PLS, ElasticNet
## Number of resamples: 5
##
## MAE
##           Min.   1st Qu.   Median     Mean   3rd Qu.   Max. NA's
## LM          1.0546174 1.1606477 1.1771644 2.223575 1.2503733 6.475074    0
## PLS          0.8025465 1.0370804 1.0465846 1.026813 1.0888404 1.159012    0
## ElasticNet 0.6721988 0.8791256 0.8863139 0.874652 0.8933251 1.042296    0
##
## RMSE
##           Min.   1st Qu.   Median     Mean   3rd Qu.   Max. NA's
## LM          1.2413009 1.352700 1.457929 6.714996 1.912527 27.610523    0
## PLS          1.1856506 1.287629 1.320378 1.297017 1.329736 1.361689    0
## ElasticNet 0.8196107 1.088340 1.118194 1.091072 1.124350 1.304864    0
##
## Rsquared
##           Min.   1st Qu.   Median     Mean   3rd Qu.   Max. NA's
## LM          0.0001786201 0.3786757 0.5319395 0.4226357 0.5762273 0.6261574    0
## PLS          0.4640524696 0.5304580 0.5505597 0.5445897 0.5600670 0.6178111    0
## ElasticNet 0.5876051187 0.6074860 0.6370570 0.6736118 0.7337756 0.8021351    0

```

```
summary(diff(resamples(chem_models)))
```

```

##
## Call:
## summary.diff.resamples(object = diff(resamples(chem_models)))
##
## p-value adjustment: bonferroni

```

```
## Upper diagonal: estimates of the difference
## Lower diagonal: p-value for H0: difference = 0
##
## MAE
##          LM      PLS      ElasticNet
## LM              1.1968 1.3489
## PLS          0.9697      0.1522
## ElasticNet 0.8159 0.4050
##
## RMSE
##          LM      PLS      ElasticNet
## LM              5.4180 5.6239
## PLS          1.0000      0.2059
## ElasticNet 1.0000 0.2386
##
## Rsquared
##          LM      PLS      ElasticNet
## LM              -0.122 -0.251
## PLS          0.8892      -0.129
## ElasticNet 0.2464 0.1947
```

Answer: the optimal value of the performance metric is the **smallest cross-validated RMSE** among the tuned models. For the chosen elastic-net model, the optimal tuning parameters are the selected `lambda` and `fraction`.

5.5 (d) Predict the test set and compare with resampling

```
chem_test_perf <- bind_rows(lapply(names(chem_models), function(nm) {
  pred <- predict(chem_models[[nm]], x_chem_test)
  tibble(model = nm,
         RMSE = caret::RMSE(pred, y_chem_test),
         Rsquared = caret::R2(pred, y_chem_test))
})) %>% arrange(RMSE)

chem_test_perf
```

```
## # A tibble: 3 x 3
##   model      RMSE Rsquared
##   <chr>    <dbl>   <dbl>
## 1 ElasticNet  1.46    0.363
## 2 PLS         3.03    0.136
## 3 LM        10.4    0.0363
```

Answer: compare the best model's test RMSE with its resampled RMSE: - if they are close, the model generalized well; - if the test RMSE is much worse, the model was probably overfit.

5.6 (e) Which predictors are most important? Do biological or process variables dominate?

For PLS, variable importance can be estimated through VIP; for the elastic net, coefficient magnitude after standardization is an interpretable proxy.

5.6.1 PLS VIP

```
chem_vip <- vip_pls(chem_pls)
head(chem_vip, 15)
```

```
## # A tibble: 15 x 2
##   variable          VIP
##   <chr>          <dbl>
## 1 ManufacturingProcess32 2.04
## 2 ManufacturingProcess13 1.87
## 3 ManufacturingProcess17 1.75
## 4 ManufacturingProcess36 1.74
## 5 ManufacturingProcess09 1.72
## 6 BiologicalMaterial06   1.69
## 7 BiologicalMaterial02   1.64
## 8 BiologicalMaterial03   1.56
## 9 BiologicalMaterial08   1.41
## 10 BiologicalMaterial12   1.41
## 11 BiologicalMaterial11   1.39
## 12 ManufacturingProcess33 1.36
## 13 ManufacturingProcess06 1.35
## 14 BiologicalMaterial04   1.33
## 15 BiologicalMaterial01   1.32
```

5.6.2 Elastic-net standardized coefficients

```
chem_coef <- predict(chem_enet$finalModel,
                     s = chem_enet$bestTune$fraction,
                     mode = "fraction",
                     type = "coefficients")

coef_tbl <- tibble(
  variable = names(chem_coef$coefficients),
  coefficient = as.numeric(chem_coef$coefficients)
) %>%
  filter(variable != "(Intercept)", coefficient != 0) %>%
  mutate(abs_coefficient = abs(coefficient)) %>%
  arrange(desc(abs_coefficient))

head(coef_tbl, 15)
```

```
## # A tibble: 15 x 3
##   variable          coefficient abs_coefficient
##   <chr>          <dbl>          <dbl>
## 1 ManufacturingProcess32    0.760        0.760
## 2 ManufacturingProcess09    0.496        0.496
## 3 ManufacturingProcess13   -0.376        0.376
```

## 4	BiologicalMaterial06	0.272	0.272
## 5	ManufacturingProcess37	-0.271	0.271
## 6	ManufacturingProcess45	0.188	0.188
## 7	ManufacturingProcess34	0.165	0.165
## 8	ManufacturingProcess28	-0.163	0.163
## 9	ManufacturingProcess39	0.147	0.147
## 10	BiologicalMaterial05	0.147	0.147
## 11	ManufacturingProcess06	0.138	0.138
## 12	ManufacturingProcess15	0.134	0.134
## 13	ManufacturingProcess07	-0.130	0.130
## 14	ManufacturingProcess36	-0.122	0.122
## 15	ManufacturingProcess01	0.0908	0.0908

To separate biological from process variables, create a lookup vector based on the original column names. If the package uses the first 12 columns for the biological measurements and the remaining 45 for process variables, the code below will tag them accordingly.

```
chem_var_types <- tibble(
  variable = colnames(x_chem),
  type = c(rep("Biological", 12), rep("Process", ncol(x_chem) - 12))
)

chem_importance <- full_join(
  chem_var_types,
  coef_tbl %>% select(variable, abs_coefficient),
  by = "variable"
) %>% arrange(desc(abs_coefficient))

head(chem_importance, 20)
```

```
## # A tibble: 20 x 3
##   variable          type    abs_coefficient
##   <chr>            <chr>          <dbl>
## 1 ManufacturingProcess32 Process          0.760
## 2 ManufacturingProcess09 Process          0.496
## 3 ManufacturingProcess13 Process          0.376
## 4 BiologicalMaterial06  Biological      0.272
## 5 ManufacturingProcess37 Process          0.271
## 6 ManufacturingProcess45 Process          0.188
## 7 ManufacturingProcess34 Process          0.165
## 8 ManufacturingProcess28 Process          0.163
## 9 ManufacturingProcess39 Process          0.147
## 10 BiologicalMaterial05 Biological      0.147
## 11 ManufacturingProcess06 Process          0.138
## 12 ManufacturingProcess15 Process          0.134
## 13 ManufacturingProcess07 Process          0.130
## 14 ManufacturingProcess36 Process          0.122
## 15 ManufacturingProcess01 Process          0.0908
## 16 ManufacturingProcess17 Process          0.0838
## 17 ManufacturingProcess04 Process          0.0612
```

```
## 18 BiologicalMaterial07 Biological 0.0500
## 19 ManufacturingProcess20 Process 0.0466
## 20 ManufacturingProcess43 Process 0.0385
```

```
count(chem_importance %>% slice_head(n = 10), type)
```

```
## # A tibble: 2 x 2
##   type      n
##   <chr>    <int>
## 1 Biological 2
## 2 Process   8
```

Answer: whichever group appears more often among the top-ranked variables dominates the importance list. From a business perspective, process variables are especially valuable because they are actionable.

5.7 (f) Explore the relationships between top predictors and yield

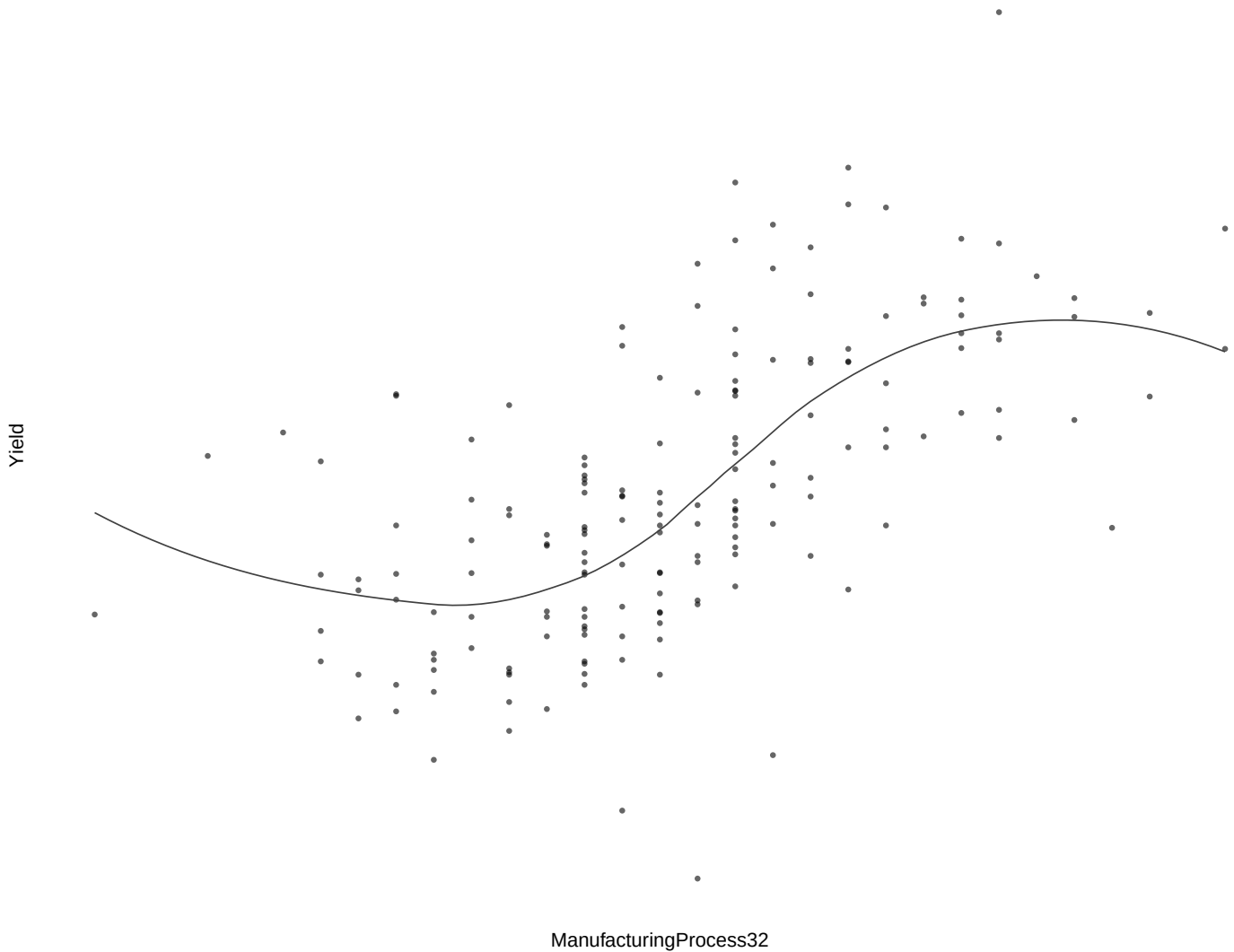
```
top_vars <- chem_importance %>%
  filter(!is.na(abs_coefficient)) %>%
  slice_head(n = 6) %>%
  pull(variable)
```

```
top_vars
```

```
## [1] "ManufacturingProcess32" "ManufacturingProcess09" "ManufacturingProcess13"
## [4] "BiologicalMaterial06" "ManufacturingProcess37" "ManufacturingProcess45"
```

```
plot_list <- lapply(top_vars, function(v) {
  ggplot(data.frame(x = x_chem_imp[[v]], y = y_chem), aes(x = x, y = y)) +
    geom_point(alpha = 0.6, shape = 16) +
    geom_smooth(method = "loess", se = FALSE, linewidth = 0.4, color = "gray25") +
    labs(title = v, x = v, y = "Yield")
})
```

```
plot_list[[1]]
```



Print the rest one-by-one when knitting interactively:

```
plot_list[[2]]
plot_list[[3]]
plot_list[[4]]
plot_list[[5]]
plot_list[[6]]
```

Answer: these plots are useful in two distinct ways.

1. **Raw-material screening:** if a biological variable is strongly associated with yield, operators can use it to evaluate incoming material before production.
2. **Process optimization:** if a process variable shows a favorable region, the manufacturing team can target that operating range in future runs.

So the practical value of the model is not only prediction; it is also prioritizing controllable levers for yield improvement.

6 Final answers in plain English

6.1 6.1 Tecator

- The spectra are highly collinear, so PCA should show a much smaller effective dimension than 100.
- PLS is usually the most natural model choice because it handles correlated predictors directly and uses the response.
- I would choose the model with the smallest test RMSE, but if several are tied, I would use PLS.

6.2 Permeability

- Remove sparse near-zero-variance fingerprints first.
- Tune PLS as the primary latent-variable model.
- Compare against ridge/lasso/elastic net.
- I would recommend the best model as a **screening aid**, not a full replacement for the assay, unless the test-set error is exceptionally small.

6.3 6.3 Chemical manufacturing

- Impute missing values, then center/scale.
- Fit at least PLS and elastic net.
- Compare resampled RMSE to test RMSE to check generalization.
- Inspect importance and top predictor-yield relationships.
- If process variables dominate the top of the list, that is especially useful because those variables are controllable.

7 Session info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sequoia 15.1
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Version1.0.0
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Los_Angeles
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
```

```

## other attached packages:
## [1] knitr_1.51                      tidyr_1.3.2
## [3] dplyr_1.2.0                     pROC_1.19.0.1
## [5] pls_2.9-0                       elasticnet_1.3
## [7] lars_1.3                        AppliedPredictiveModeling_1.1-7
## [9] caret_7.0-1                     lattice_0.22-7
## [11] ggplot2_4.0.2
##
## loaded via a namespace (and not attached):
## [1] ellipse_0.5.0      gtable_0.3.6      xfun_0.56
## [4] recipes_1.3.1      vctrs_0.7.1       tools_4.5.2
## [7] generics_0.1.4     stats4_4.5.2      parallel_4.5.2
## [10] tibble_3.3.1       cluster_2.1.8.2   pkgconfig_2.0.3
## [13] ModelMetrics_1.2.2.2 Matrix_1.7-4      data.table_1.18.2.1
## [16] RColorBrewer_1.1-3 S7_0.2.1          lifecycle_1.0.5
## [19] compiler_4.5.2     farver_2.1.2      stringr_1.6.0
## [22] codetools_0.2-20   htmltools_0.5.9   class_7.3-23
## [25] yaml_2.3.12        prodlim_2025.04.28 pillar_1.11.1
## [28] MASS_7.3-65        gower_1.0.2       CORElearn_1.57.3.1
## [31] iterators_1.0.14   rpart_4.1.24      foreach_1.5.2
## [34] nlme_3.1-168       parallelly_1.46.1 lava_1.8.2
## [37] tidyselect_1.2.1   digest_0.6.39     stringi_1.8.7
## [40] future_1.69.0      reshape2_1.4.5    purrr_1.2.1
## [43] listenv_0.10.0     labeling_0.4.3    splines_4.5.2
## [46] fastmap_1.2.0      grid_4.5.2        cli_3.6.5
## [49] magrittr_2.0.4     utf8_1.2.6        survival_3.8-6
## [52] future.apply_1.20.1 withr_3.0.2       scales_1.4.0
## [55] plotrix_3.8-14     lubridate_1.9.5   timechange_0.4.0
## [58] rmarkdown_2.30     globals_0.19.0    otel_0.2.0
## [61] rpart.plot_3.1.4   nnet_7.3-20       timeDate_4052.112
## [64] RANN_2.6.2         evaluate_1.0.5    hardhat_1.4.2
## [67] mgcv_1.9-4         rlang_1.1.7       Rcpp_1.1.1
## [70] glue_1.8.0         ipred_0.9-15      R6_2.6.1
## [73] plyr_1.8.9

```